No.

Date:

Bagian I

1. ubahlah ukuran sudut berikut ke bentuk desimal !

a. $21°45'30'' = \frac{30}{60} = 0,5 + 45 = \frac{45,5}{60} = 0,7583 + 21 = 21,7583$

b. $45°15' = \frac{15}{60} = 0,25 + 45 = 45,25$

c. $61°42'21'' = \frac{21}{60} = 0,35 + 42 = \frac{42,35}{60} = 0,70583 + 61 = 61,70583$

d. $37°11'29'' = \frac{29}{60} = 0,483 + 11 = \frac{11,483}{60} = 0,19138 + 37 = 37,19138$

e. $61°42'21'' = \frac{21}{60} = 0,35 + 42 = \frac{42,35}{60} = 0,70583 + 61 = 61,70583$

2. ubahlah ukuran sudut berikut ke benduk $D°M'S''$!

a. $61.24° = 61°14'24''$

b. $28.6582° = 28°39'29,52''$

c. $165.0525° = 165°51'9''$

d. $225,24° = 225°14'24''$

e. $-16.24° = -16°14'24''$

3. ubahlah ukuran sudut berikut ke bentuk radian !

a. $30° = \frac{30}{180} = \frac{1}{6} = \frac{\pi}{6}$ rad

b. $60° = \frac{60}{180} = \frac{1}{3} = \frac{\pi}{3}$ rad

c. $135° = \frac{135}{180} = \frac{3}{4}\pi$ rad

d). $240° = \frac{240}{180} = \frac{4}{3}\pi$ rad

e). $-60° = \frac{-60}{180} = -\frac{1}{3}\pi = -\frac{\pi}{3}$ rad

Bagian II

1. $8 \sin x = 4$

$\sin x = \frac{4}{8} = \frac{1}{2}$

$\sin x = \frac{1}{2}$

$x = 30°, 150°$

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3. $20 \sin x + 10\sqrt{3} = 0$

$20 \sin x = -10\sqrt{3}$

$\sin x = \frac{-10\sqrt{3}}{20}$

$\sin x = \frac{-1\sqrt{3}}{2} = \frac{-\sqrt{3}}{2} = 60^\circ$

$(180 + \alpha) = (180 + 60) = 240^\circ$

$(360 - \alpha) = (360 - 60) = 300^\circ$

5. $10 \sin (x + 30^\circ) = 5$

$\sin (x + 30^\circ) = \frac{5}{10} = \frac{1}{2}$

$\sin (x + 30^\circ) = \frac{1}{2}$

$\sin (x + 30^\circ) = \sin 30$

- $\sin (x+30^\circ) = \sin 30^\circ$

$x + 30 = (180^\circ - 30^\circ)$

$k = 0 = x + 30 = 150^\circ + 0$

$x = 150^\circ - 30^\circ = 120^\circ$

- $\sin (x + 30^\circ) = \frac{1}{2}$

$\sin (x + 30^\circ) = \sin 30^\circ$

$x + 30^\circ = 30 + k.360$

$k = 0 = x + 30 = 30 + 0$

$x = 30 + 0 - 30 = 0^\circ$

→ $k = 1 = x + 30 = 30 + 360$

$x = 30 + 360 - 30$

$= 360^\circ$

Jadi $\{0^\circ, 120^\circ, 360^\circ\}$

selesaikan persamaan trigonometri berikut dgn x antara 0 dan 2π radian

7. $12 \cos (x - \frac{\pi}{4}) = 6$

$\cos \left(x - \frac{\pi}{4} \right) = \frac{6}{12} = \frac{1}{2}$

$\cos \theta = \frac{1}{2}$, nilai θ dari 0 hingga $2\pi = \frac{\pi}{3}$ dan $\frac{5\pi}{3}$

mencari nilai x dan $\frac{\pi}{3}$ dan $\frac{5\pi}{3}$

$x = \frac{\pi}{3} + \frac{\pi}{4} = \frac{4\pi}{12} + \frac{3\pi}{12} = \frac{7\pi}{12}$

$x = \frac{5\pi}{3} + \frac{\pi}{4} = \frac{20\pi}{12} + \frac{3\pi}{12} = \frac{23\pi}{12}$

9. $4\sin^2 x - 3 = 0$

$4\sin^2 x = 3$

$\sin^2 x = \frac{3}{4} = \pm\frac{\sqrt{3}}{4} \cdot \frac{\sqrt{3}}{2}$

$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$